Granular Instrumental Variables

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Aggregate shocks confound firm effects

The Goal: Estimate the causal elasticity of wages to labor demand.

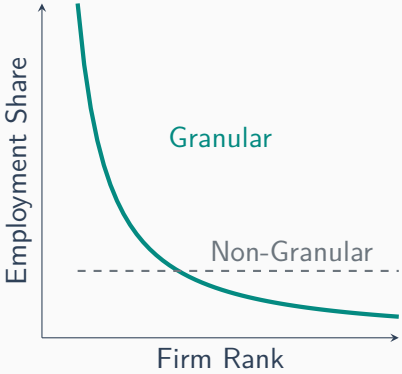
The Identification Challenge: **Simultaneity**

- **Supply Equation:** Wages respond to labor demand and macro supply shocks.
- **Demand Equation:** Labor demand responds to wages and macro demand shocks.

LLN fails in granular economies

1. In a standard economy, micro-shocks cancel out at a rate of $1/\sqrt{N}$ ▸ Proof

2. In a **granular** economy, where the size distribution of firms follows a power law (i.e., Zipf's Law: size $\propto \frac{1}{\text{rank}}$), a few firms are so large that their idiosyncratic luck drives aggregate volatility.



A Simple Model

Simultaneity prevents OLS estimation

Consider an economy with N firms. The system is given by:

$$w_t = \psi L_{St} + \varepsilon_t \tag{Supply}$$

$$L_{it} = \phi^d w_t + \eta_t + u_{it} \tag{Demand}$$

Where:

- w_t is the wage (observed)
- L_{it} is the number of workers at firm i (observed)
- η_t and ε_t are aggregate shocks (unobserved)
- u_{it} is the idiosyncratic shock to firm i (unobserved)
- Main assumption: u_{it} are uncorrelated with aggregate shocks (η_t, ϵ_t) Assumptions

We cannot estimate ψ (labor supply elasticity) and ϕ^d (labor demand elasticity) by OLS, due to simultaneity.

GIV differences size and equal weights

The GIV exploits the difference between size-weighted and equal-weighted averages.

Define the equal-weighted average outcome:

$$L_{Et} = \frac{1}{N} \sum_{i=1}^{N} L_{it} \qquad \text{(Equal-Weighted Average)}$$

Define the size-weighted average outcome:

$$L_{St} = \sum_{i=1}^{N} S_i L_{it}, \quad \sum_{i=1}^{N} S_i = 1 \qquad \text{(Size-Weighted Average)}$$

The GIV is the size-weighted outcome minus the equal-weighted outcome:

$$z_t := L_{\Gamma t} = L_{St} - L_{Et} \qquad \text{(GIV)}$$

GIV isolates idiosyncratic shocks

Substitute the demand equation into the aggregate definitions: Proof

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$
$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

The common components $(\phi^d w_t + \eta_t)$ difference out:

$$z_t = u_{\Gamma t} = u_{St} - u_{Et} = \sum_{i=1}^{N} S_i u_{it} - \frac{1}{N}\sum_{i=1}^{N} u_{it} = \sum_{i=1}^{N}(S_i - \frac{1}{N})u_{it} \qquad \text{(GIV)}$$

z_t is made *only* of idiosyncratic shocks.

GIV enables 2SLS estimation

- **First Stage:** Regress aggregate labor demand (L_{St}) on the granular instrument (z_t). This extracts $\widehat{L}_{St}$, the variation driven *only* by the idiosyncratic shocks of firms.

- **Second Stage:** Regress aggregate wages (w_t) on the predicted labor demand from the First Stage ($\widehat{L}_{St}$) to obtain an unbiased estimate of ψ.

GIV requires relevance and exclusion

For z_t to be a valid instrument, two fundamental assumptions must hold in a granular economy.

- **Relevance (Cov$(L_{St}, z_t) \neq 0$):**

  The granular shocks must actually move the aggregate market. This requires a *concentrated market* (e.g., high Herfindahl index) where the idiosyncratic shocks of the largest firms are volatile enough to shift total labor demand.

- **Exclusion Restriction (Cov$(z_t, \varepsilon_t) = 0$, Cov$(z_t, \eta_t) = 0$):**

  The idiosyncratic shocks of large firms (z_t) must not directly affect aggregate wages (w_t) through any channel other than aggregate labor demand (L_{St}). They must be truly orthogonal to macro aggregate supply shocks ($z_t \perp \eta_t, \varepsilon_t$).

GIV yields both elasticities

By exogeneity, $\mathbb{E}[u_t \varepsilon_t] = 0 \implies \mathbb{E}[z_t \varepsilon_t] = 0$, where $u_t = (u_{it})_{i=1,\ldots,N}$.

Using the supply equation:

$$w_t = \psi L_{St} + \varepsilon_t \qquad \implies \qquad \mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$$

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]} \qquad \text{(Supply Elasticity)}$$

Proof

Using the equal-weighted demand equation, $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$:

$$\phi^d = \frac{\mathbb{E}[L_{Et} z_t]}{\mathbb{E}[w_t z_t]} \qquad \text{(Demand Elasticity)}$$

The Full Model

Handling heterogeneous factor loadings

Heterogeneous loadings bias simple GIV

We now allow for heterogeneous loadings λ_i:

$$L_{it} = \phi w_t + \lambda_i \eta_t + u_{it}$$

The simple GIV $(L_{St} - L_{Et})$ assumes every firm has the same loading $\lambda_i = 1$. If large firms are more sensitive to the Fed, the simple GIV is **"leaky"**—it correlates with η_t.

The simple GIV becomes: Proof

$$z_t = (\lambda_S - \lambda_E)\eta_t + (u_{St} - u_{Et})$$

If $\lambda_S \neq \lambda_E$, the instrument is **invalid** ($\mathbb{E}[z_t \eta_t] \neq 0$, i.e., the Exclusion Restriction fails).

Orthogonal weights restore exclusion

Substitute Equation $L_{it} = \phi w_t + \lambda_i \eta_t + u_{it}$ into the definition of the weighted GIV:

$$\begin{aligned} z_t &= \sum \Gamma_i L_{it} = \sum \Gamma_i (\phi w_t + \lambda_i \eta_t + u_{it}) \\ &= \phi w_t (\sum \Gamma_i) + \eta_t (\sum \Gamma_i \lambda_i) + \sum \Gamma_i u_{it} \end{aligned}$$

Where Γ is the vector of weights that we want to find.
To ensure z_t contains *only* idiosyncratic shocks (u_{it}), we must satisfy two conditions:

1. $\sum \Gamma_i = 0$: This removes the endogenous wage w_t.
2. $\sum \Gamma_i \lambda_i = 0$: This removes the common macro shock η_t.

We need weights Γ such that the factor components **cancel exactly**. This ensures $z_t = \sum \Gamma_i u_{it}$ is purely idiosyncratic.

Projection finds optimal weights

For this, we use the $N \times N$ projection matrix Q that is orthogonal to the factor loadings—that is, $Q\lambda = 0$: ▸ Refresher

$$Q = I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

The optimal weights are then given by $\Gamma^{*\prime} = S'Q$. ▸ Propositions

- Intuitively, we want Γ to be as close as possible to S but satisfy $\Gamma \perp \lambda$. This is a standard projection problem: $\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda' S$
- In essence, we take the firm sizes and subtract the portion of size that is explained by macro-sensitivity. What remains is pure granular weight.

Application

GIV in the Israeli Labor Market

Israeli tech is highly granular

The Israeli high-tech industry is dominated by a few massive multinational R&D centers.

- **High Concentration:** Companies like Intel, Google, Microsoft, and NVIDIA account for a substantial, disproportionate share of tech employment.
- **A Heavy Tail:** The size distribution of tech employers in Israel exhibits a strong power law, yielding a high **excess Herfindahl index (h)**.
- **Macro vs. Micro:** It is difficult to separate general Israeli macroeconomic conditions (e.g., security shocks, interest rates) from the hiring dynamics of these specific giants.

GIV unconfounds tech spillovers

How do we isolate the causal spillover of these tech giants on aggregate Israeli wages?

- **The Idiosyncratic Shocks (u_{it}):** We exploit global shocks specific to the multinationals that are arguably exogenous to the Israeli macroeconomy (e.g., global restructuring at Intel, a worldwide AI hiring boom at NVIDIA).
- **The Instrument (z_t):** Construct the GIV by taking the size-weighted employment shocks of these giants and projecting out the common Israeli macroeconomic factor loadings.
- **The 2SLS Estimation:** Regress aggregate Israeli wages on the component of labor demand driven *only* by these isolated, granular tech shocks.

GIV unconfounds macro dynamics

The GIV method solves the simultaneity problem in concentrated labor markets.

- **Causal Wage Spillovers:** We can measure exactly how much a hiring freeze at a major tech firm depresses aggregate local wages, separated from a general economic downturn.
- **Beyond the US:** The method is perfectly suited for small, open economies like Israel, where granular actors often drive aggregate volatility.

Appendix A

Derivations

Assumptions

The Central Identification Assumption

Individual shocks u_{it} are truly idiosyncratic—uncorrelated with aggregate shocks (η_t, ϵ_t) and controls (C_t^y, C_t^p):

$$u_t \perp \eta_t, \epsilon_t, C_t^y, C_t^p \tag{1}$$

Baseline Simplifications

A1 Factor loadings λ_i are known (relaxed in §IV.B).

A2 Controls have factor structure: $C_{it}^y = \lambda_i c_t^y$.

Error Structure

A3 Homoskedastic and uncorrelated across i: $\mathbb{E}[u_t u_t'] = \sigma_u^2 I$.

A4 General heteroskedasticity V^u (relaxed for optimality).

Note: All random variables are assumed to have finite fourth moments and are i.i.d. across periods.

Back

Equal weights decay at $1/\sqrt{N}$

Suppose there are N firms, each receiving an independent idiosyncratic shock u_{it}. Assume these shocks have mean 0 and a common variance σ^2:

$$\mathbb{E}[u_{it}] = 0, \quad \text{Var}(u_{it}) = \sigma^2, \quad \text{Cov}(u_{it}, u_{jt}) = 0 \text{ for } i \neq j$$

In a non-granular economy, the aggregate shock is an **equal-weighted** average:

$$u_{Et} = \frac{1}{N} \sum_{i=1}^{N} u_{it}$$

The variance of the aggregate shock is:

$$\text{Var}(u_{Et}) = \text{Var}\left(\frac{1}{N} \sum u_{it}\right) = \frac{1}{N^2} \sum \text{Var}(u_{it}) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$$

The aggregate volatility is therefore: $\sigma_{agg} = \frac{\sigma}{\sqrt{N}}$

Back

Size-weighting aggregates shocks

The size-weighted outcome is defined as $L_{St} = \sum_{i=1}^{N} S_i L_{it}$, where S_i are weights such that $\sum S_i = 1$.

$$L_{St} = \sum_{i=1}^{N} S_i(\phi^d w_t + \eta_t + u_{it})$$

$$L_{St} = \sum_{i=1}^{N} S_i \phi^d w_t + \sum_{i=1}^{N} S_i \eta_t + \sum_{i=1}^{N} S_i u_{it}$$

Since ϕ^d, w_t, and η_t do not depend on the index i, we can pull them out:

$$L_{St} = \phi^d w_t \left(\sum_{i=1}^{N} S_i \right) + \eta_t \left(\sum_{i=1}^{N} S_i \right) + \sum_{i=1}^{N} S_i u_{it}$$

Using the property $\sum S_i = 1$ and defining the size-weighted idiosyncratic shock as $u_{St} = \sum S_i u_{it}$, we get:

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$

Back

Equal-weighting averages shocks

The equal-weighted outcome is defined as $L_{Et} = \frac{1}{N}\sum_{i=1}^{N} L_{it}$.

$$L_{Et} = \frac{1}{N}\sum_{i=1}^{N}(\phi^d w_t + \eta_t + u_{it})$$

$$L_{Et} = \frac{1}{N}\sum_{i=1}^{N}\phi^d w_t + \frac{1}{N}\sum_{i=1}^{N}\eta_t + \frac{1}{N}\sum_{i=1}^{N}u_{it}$$

Again, pulling constants out of the summations:

$$L_{Et} = \phi^d w_t\left(\frac{1}{N}\sum_{i=1}^{N}1\right) + \eta_t\left(\frac{1}{N}\sum_{i=1}^{N}1\right) + \frac{1}{N}\sum_{i=1}^{N}u_{it}$$

Since $\frac{1}{N}\sum_{i=1}^{N}1 = \frac{1}{N}\cdot N = 1$, and defining the equal-weighted idiosyncratic shock as $u_{Et} = \frac{1}{N}\sum u_{it}$, we get:

$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

GIV identifies supply elasticity

To find ψ, we use z_t as an instrument for L_{St} in the supply equation. We take the expectation of the product of the supply equation and the instrument:

$$\mathbb{E}[w_t z_t] = \mathbb{E}[(\psi L_{St} + \epsilon_t) z_t]$$
$$\mathbb{E}[w_t z_t] = \psi \mathbb{E}[L_{St} z_t] + \mathbb{E}[\epsilon_t z_t]$$

By the exogeneity condition, $\mathbb{E}[\epsilon_t z_t] = 0$. Thus:

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]} \tag{2}$$

This is the standard IV formula where z_t instruments for L_{St}.

Back

GIV identifies demand elasticity

To find ϕ^d, we use the equal-weighted aggregate demand equation derived on slide 8:

$$L_{Et} = \phi^d w_t + \eta_t + u_{Et} \tag{41}$$

We aim to show that $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$. Substituting the demand equation:

$$\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = \mathbb{E}[(\eta_t + u_{Et}) z_t] = \mathbb{E}[\eta_t z_t] + \mathbb{E}[u_{Et} z_t] \tag{3}$$

We already know $\mathbb{E}[\eta_t z_t] = 0$. Now we examine $\mathbb{E}[u_{Et} z_t]$. Under the assumption that u_{it} are i.i.d. with variance σ_u^2:

$$\begin{aligned} \mathbb{E}[u_{Et} z_t] &= \mathbb{E}[u_{Et}(u_{St} - u_{Et})] \\ &= \mathbb{E}[u_{Et} u_{St}] - \mathbb{E}[u_{Et}^2] \end{aligned}$$

Back

GIV identifies demand elasticity

$$\mathbb{E}[u_{Et}u_{St}] - \mathbb{E}[u_{Et}^2] =$$

$$= \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)\left(\sum_j S_j u_{jt}\right)\right] - \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)^2\right]$$

$$= \frac{1}{N}\sum_i S_i\sigma_u^2 - \frac{1}{N^2}\sum_i \sigma_u^2$$

$$= \frac{1}{N}\sigma_u^2(1) - \frac{1}{N}\sigma_u^2 = 0$$

Since $\mathbb{E}[u_{Et}z_t] = 0$, it follows that $\mathbb{E}[(L_{Et} - \phi^d w_t)z_t] = 0$. Rearranging:

$$\mathbb{E}[L_{Et}z_t] = \phi^d\mathbb{E}[w_t z_t]$$

$$\phi^d = \frac{\mathbb{E}[L_{Et}z_t]}{\mathbb{E}[w_t z_t]}$$

Simple GIV leaks macro shocks

The simple GIV is defined as $z_t^{simple} = L_{St} - L_{Et}$. Let's derive its components:

1. **Size-weighted average (L_{St}):**

$$\begin{aligned} L_{St} &= \sum S_i L_{it} = \sum S_i(\phi w_t + \lambda_i \eta_t + u_{it}) \\ &= \phi w_t \underbrace{\sum S_i}_{1} + \eta_t \underbrace{\sum S_i \lambda_i}_{\lambda_S} + \underbrace{\sum S_i u_{it}}_{u_{St}} \\ &= \phi w_t + \lambda_S \eta_t + u_{St} \end{aligned}$$

2. **Equal-weighted average (L_{Et}):**

$$\begin{aligned} L_{Et} &= \frac{1}{N} \sum L_{it} = \phi w_t + \eta_t \underbrace{\frac{1}{N} \sum \lambda_i}_{\lambda_E} + u_{Et} \\ &= \phi w_t + \lambda_E \eta_t + u_{Et} \end{aligned}$$

3. **The Resulting GIV:** Back

$$z_t^{simple} = L_{St} - L_{Et} = (\lambda_{\mathbf{S}} - \lambda_{\mathbf{E}})\eta_{\mathbf{t}} + (u_{St} - u_{Et})$$

Econometrics Refresher

OLS and the Annihilator Matrix

Standard OLS projections decompose a vector y into explained and unexplained parts:

- **Projection Matrix (P):** $P = X(X'X)^{-1}X'$. Projects y onto the column space of X. $\hat{y} = Py$.
- **Annihilator Matrix (M):** $M = I - P$. Projects y onto the space orthogonal to X.
- **Orthogonality:** $MX = 0$. This ensures the residuals $e = My$ are uncorrelated with the regressors X.
- **Residuals:** $e = y - \hat{y} = (I - P)y = My$.

Notation Map:
Slide $\lambda \equiv$ Standard X (Regressors/Factors)
Slide $Q \equiv$ Standard M (Residual Maker Matrix)
Slide $S \equiv$ Standard y (Data Vector)
Slide $\Gamma \equiv$ Standard e (Residuals/Granular Weights) Back

Appendix B

Propositions

Orthogonal weights isolate idiosyncratic shocks

The GIV z_t is constructed from observables, L_{it}, using weights $\Gamma \in \mathbb{R}^N$ orthogonal to factor loadings:

$$z_t := \Gamma' L_t = \sum_{i=1}^{N} \Gamma_i L_{it}$$

By construction, this isolates a linear combination of idiosyncratic shocks u_t:

$$z_t = \Gamma' u_t \quad \Rightarrow \quad \text{exogenous to aggregate shocks } \eta_t, \varepsilon_t$$

This allows us to construct the sample estimators:

$$\hat{\psi}_T = \frac{\sum_{t=1}^{T} w_t z_t}{\sum_{t=1}^{T} L_{St} z_t} \qquad \hat{\phi}_T^d = \frac{\sum_{t=1}^{T} L_{\tilde{E}t} z_t}{\sum_{t=1}^{T} w_t z_t}$$

Back

Moments identify elasticities

The estimators follow directly from the exogeneity of z_t to structural shocks:

Inverse Supply Elasticity

From $w_t = \psi L_{St} + \cdots + \varepsilon_t$:

$$\mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$$

$$\Rightarrow \quad \psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]}$$

Demand Elasticity

For weights $\tilde{E}$ where $\mathbb{E}[u_{\tilde{E}t} u_{\Gamma t}] = 0$:

$$\mathbb{E}[(L_{\tilde{E}t} - \phi^d w_t) z_t] = 0$$

$$\Rightarrow \quad \phi^d = \frac{\mathbb{E}[L_{\tilde{E}t} z_t]}{\mathbb{E}[w_t z_t]}$$

Back

Proposition 1

Under standard regularity conditions, as the number of time periods $T \to \infty$:

Inverse Supply Elasticity

$$\hat{\psi}_T \xrightarrow{\text{a.s.}} \psi$$

Demand Elasticity

$$\hat{\phi}^d_T \xrightarrow{\text{a.s.}} \phi^d \quad (\text{if } \psi \neq 0)$$

Intuition: The orthogonal weighting Γ ensures the instrument is uncorrelated with the structural error term, resolving simultaneity bias.

Back

Multipliers capture amplification

To understand the variance of these estimators, we define two key multipliers:

Quantity Pass-through

$$M := \frac{1}{1 - \psi\phi^d}$$

Price Pass-through

$$\mu := \psi M$$

These represent the pass-through from idiosyncratic and aggregate shocks to the aggregate quantity and the price, respectively.

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Proposition 2

The estimators converge in distribution to a normal distribution:

$$\sqrt{T}(\hat{\psi}_T - \psi) \xrightarrow{d} \mathcal{N}(0, \sigma_\psi^2), \quad \sigma_\psi = \frac{\mathbb{E}[u_{\Gamma t}^2]^{1/2}}{|\mathbb{E}[u_{St} u_{\Gamma t}]|} \frac{\sigma_{\varepsilon^\psi}}{|M|}$$

$$\sqrt{T}(\hat{\phi}_T^d - \phi^d) \xrightarrow{d} \mathcal{N}(0, \sigma_{\phi^d}^2), \quad \sigma_{\phi^d} = \frac{\mathbb{E}[u_{\Gamma t}^2]^{1/2}}{|\mathbb{E}[u_{St} u_{\Gamma t}]|} \frac{\sigma_{\varepsilon^\phi}}{|\mu|}$$

Back

Volatility improves precision

The variance equations reveal the signal-to-noise ratio inherent to GIV:

Signal Strength

$|\mathbb{E}[u_{St}u_{\Gamma t}]|$

Larger granular shocks increase relevance.

Instrument Noise

$\mathbb{E}[u_{\Gamma t}^2]^{1/2}$

Variance of the constructed GIV itself.

Structural Noise

$\sigma_\varepsilon/|M|$

Unobserved shocks dampening precision.

Back

Proposition 3

To minimize the asymptotic variances σ^2_ψ and $\sigma^2_{\phi^d}$, we use the $N \times N$ projection matrix Q that is orthogonal to the factor loadings λ:

$$Q := I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

The optimal weights vector Γ^* is the projection of the size vector S:

$$\Gamma^* = S'Q$$

Result: Γ^* is the vector closest to S, while remaining orthogonal to the unobserved aggregate factor loadings.